> **PROMPT**

Consider $N = 7$ nonlinear optical cavity that connects in a ring one by one. Neglect the dissipation of the cavity, the Hamiltonian is:

$$H = \sum_{i=1}^{N} \left(\omega a_i^\dagger a_i + \frac{1}{2} U a_i^\dagger a_i^\dagger a_i a_i - J a_i^\dagger a_{i+1} - J a_i^\dagger a_{i-1} \right)$$

and $J \geq 0$. Consider 4 photons in the cavity, calculate the ground state energy when $U \to \infty$.

PREFILLED BEGINING OF OPUS REASONING `We have a Bose-Hubbard ring with N=7 sites, 4 photons, U→∞ (hard-core bosons).`

Opus 4.8

DECODED REASONING

We have a Bose-Hubbard ring with N=7 sites, 4 photons, U→∞ (hard-core bosons). We need ground state energy. With U→∞, particles behave as hard-core bosons. On a 1D ring, hard-core bosons can be mapped to free fermions via Jordan-Wigner. But on a ring there's a subtlety with boundary conditions depending on particle number parity. [···]

Kimi-K3

REASONING — PREFILLED REASONING

We have a Bose-Hubbard ring with N=7 sites, 4 photons, U→∞ (hard-core bosons). The Hamiltonian per site: ω n_i, hopping terms: -J a_i† a_{i+1} - J a_i† a_{i-1}. Note double counting: each bond appears twice, so effective hopping amplitude 2J. Hard-core bosons map to free fermions on a ring with boundary condition subtleties. Number of particles M=4 (even) → antiperiodic bound [···]

Kimi-K3

REASONING — NOT PREFILLED

We need hard-core bosons ring, 4 particles 7 sites. Hamiltonian hopping each site has -J to both neighbors, causes double counting? Expression -J a_i†a_{i+1}-J a_i†a_{i-1}; summing i double counts oriented bonds: each undirected bond represented a_i†a_j and a_j†a_i in first terms alone already, second duplicates. Standard intended Bose Hubbard H sum onsite -J sum [···]

> **RESPONSE — OPUS 4.8**

Setting Up the Problem

With $U \to \infty$, double occupancy is forbidden — the photons become **hard-core bosons**. On a 1D ring, hard-core bosons map onto **spinless free fermions** via the Jordan–Wigner transformation.

[···]

> **RESPONSE — KIMI-K3, PREFILLED**

Setting Up the Problem

With $U \to \infty$, double occupancy is forbidden: the photons become **hard-core bosons**. On a 1D ring, hard-core bosons map onto **free spinless fermions** via the Jordan–Wigner transformation.

[···]

> **RESPONSE — KIMI-K3, NOT PREFILLED**

As \(U\to\infty\), double occupation is forbidden, so the photons become hard-core bosons. On a one-dimensional ring these map to free fermions. For four particles on a seven-site ring, the fermions satisfy antiperiodic boundary conditions, giving allowed momenta

[···]